



Mobile Communications

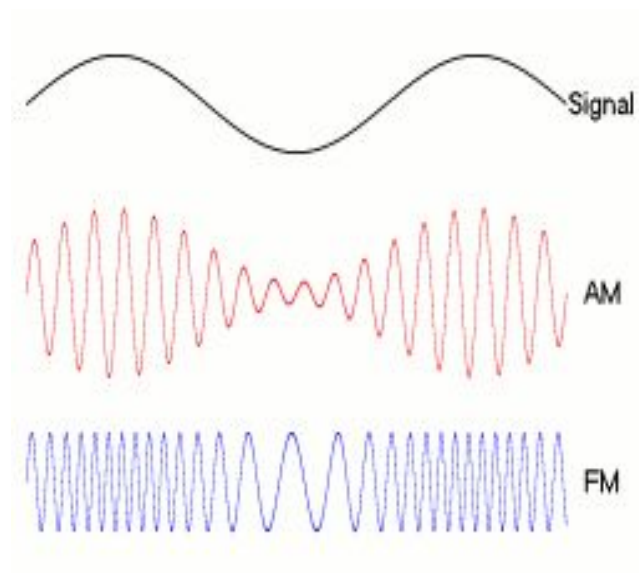
Chapter 6: Broadcast Systems

- ☐ Unidirectional distribution systems
- ☐ DAB
 - ☐ architecture
- ☐ DVB
 - ☐ Container
 - ☐ High-speed Internet





Analog Transmission





Unidirectional distribution systems

Asymmetric communication environments

- ❑ bandwidth limitations of the transmission medium
- ❑ depends on applications, type of information
- ❑ examples
 - wireless networks with basestation and mobile terminals
 - client-server environments (diskless terminal)
 - cable TV with set-top box
 - information services (pager, SMS)

Special case: unidirectional distribution systems

- ❑ high bandwidth from server to client (downstream), but no bandwidth viceversa (upstream)
- ❑ problems of unidirectional broadcast systems
 - a sender can optimize transmitted information only for one group of users/terminals
 - functions needed to individualize personal requirements/applications

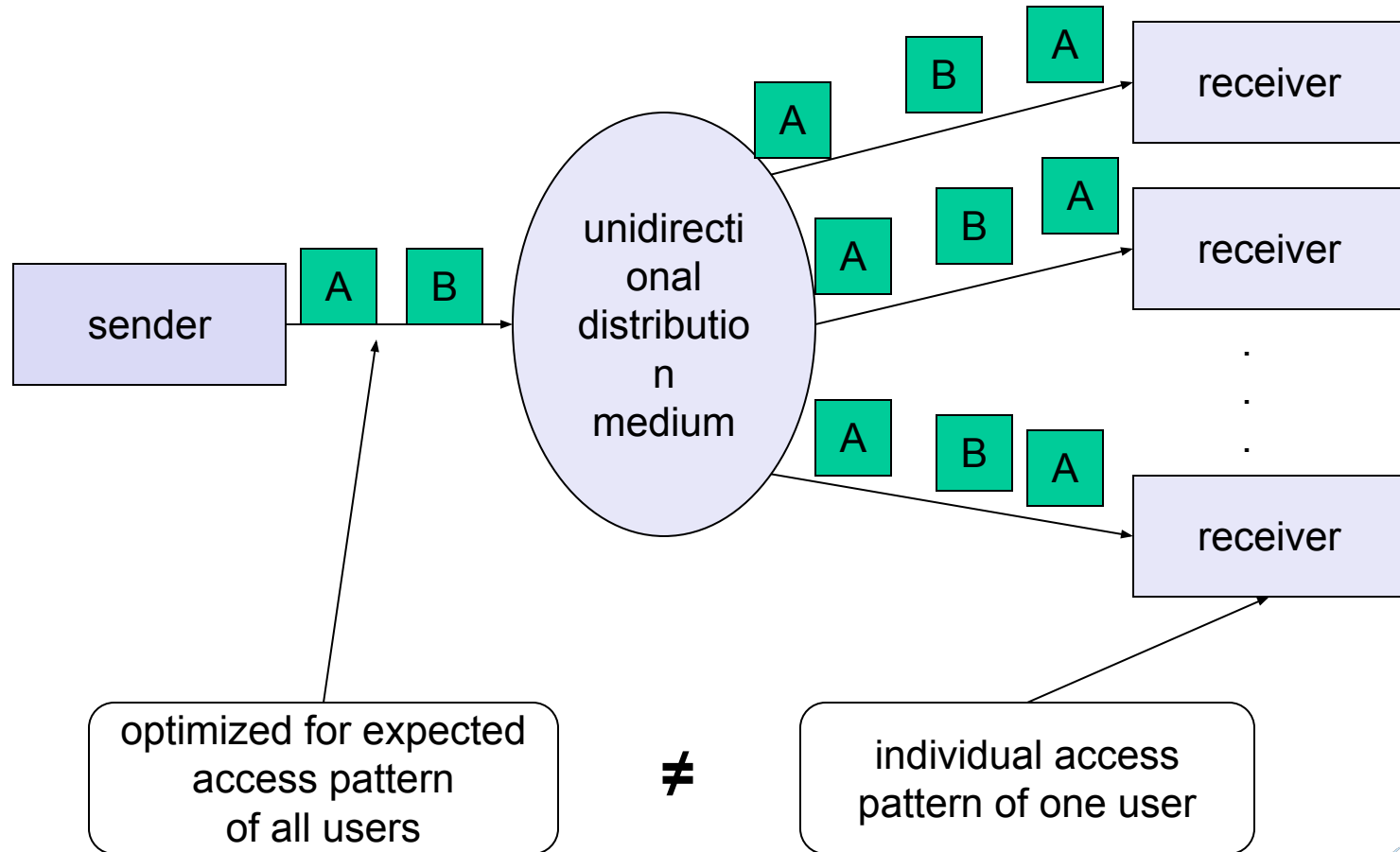




Unidirectional distribution

service provider

service user



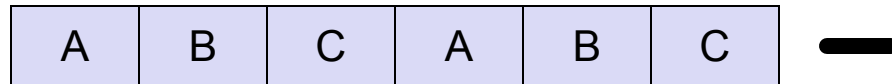


Structuring transmissions - broadcast disks

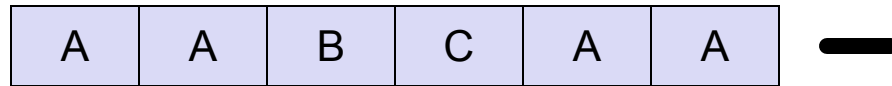
Sender

- ❑ cyclic repetition of data blocks
- ❑ different patterns possible (optimization possible only if the content is known)

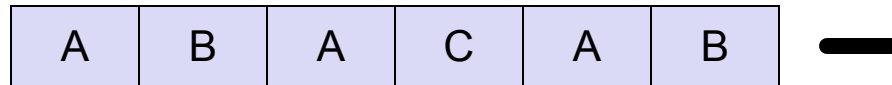
flat disk



skewed disk



multi-disk



Receiver

- ❑ use of caching
 - cost-based strategy: what are the costs for a user (waiting time) if a data block has been requested but is currently not cached
 - application and cache have to know content of data blocks and access patterns of user to optimize





DAB: Digital Audio Broadcasting

- ❑ Media access
 - ❑ COFDM (Coded Orthogonal Frequency Division Multiplex)
 - ❑ SFN (Single Frequency Network)
 - ❑ 192 to 1536 subcarriers within a 1.5 MHz frequency band
- ❑ Frequencies
 - ❑ first phase: one out of 32 frequency blocks for terrestrial TV channels 5 to 12 (174 - 230 MHz, 5A - 12D)
 - ❑ second phase: one out of 9 frequency blocks in the L-band (1452- 1467.5 MHz, LA - LI)
- ❑ Sending power: 6.1 kW (VHF, Ø 120 km) or 4 kW (L-band, Ø 30 km)
- ❑ Data-rates: 2.304 Mbit/s (net 1.2 to 1.536 Mbit/s)
- ❑ Modulation: Differential 4-phase modulation (D-QPSK)
- ❑ Audio channels per frequency block: typ. 6, max. 192 kbit/s
- ❑ Digital services: 0.6 - 16 kbit/s (PAD), 24 kbit/s (NPAD)





DAB transport mechanisms

MSC (Main Service Channel)

- ❑ carries all user data (audio, multimedia, ...)
- ❑ consists of CIF (Common Interleaved Frames)
- ❑ each CIF 55296 bit, every 24 ms (depends on transmission mode)
- ❑ CIF contains CU (Capacity Units), 64 bit each

FIC (Fast Information Channel)

- ❑ carries control information
- ❑ consists of FIB (Fast Information Block)
- ❑ each FIB 256 bit (incl. 16 bit checksum)
- ❑ defines configuration and content of MSC

Stream mode

- ❑ transparent data transmission with a fixed bit rate

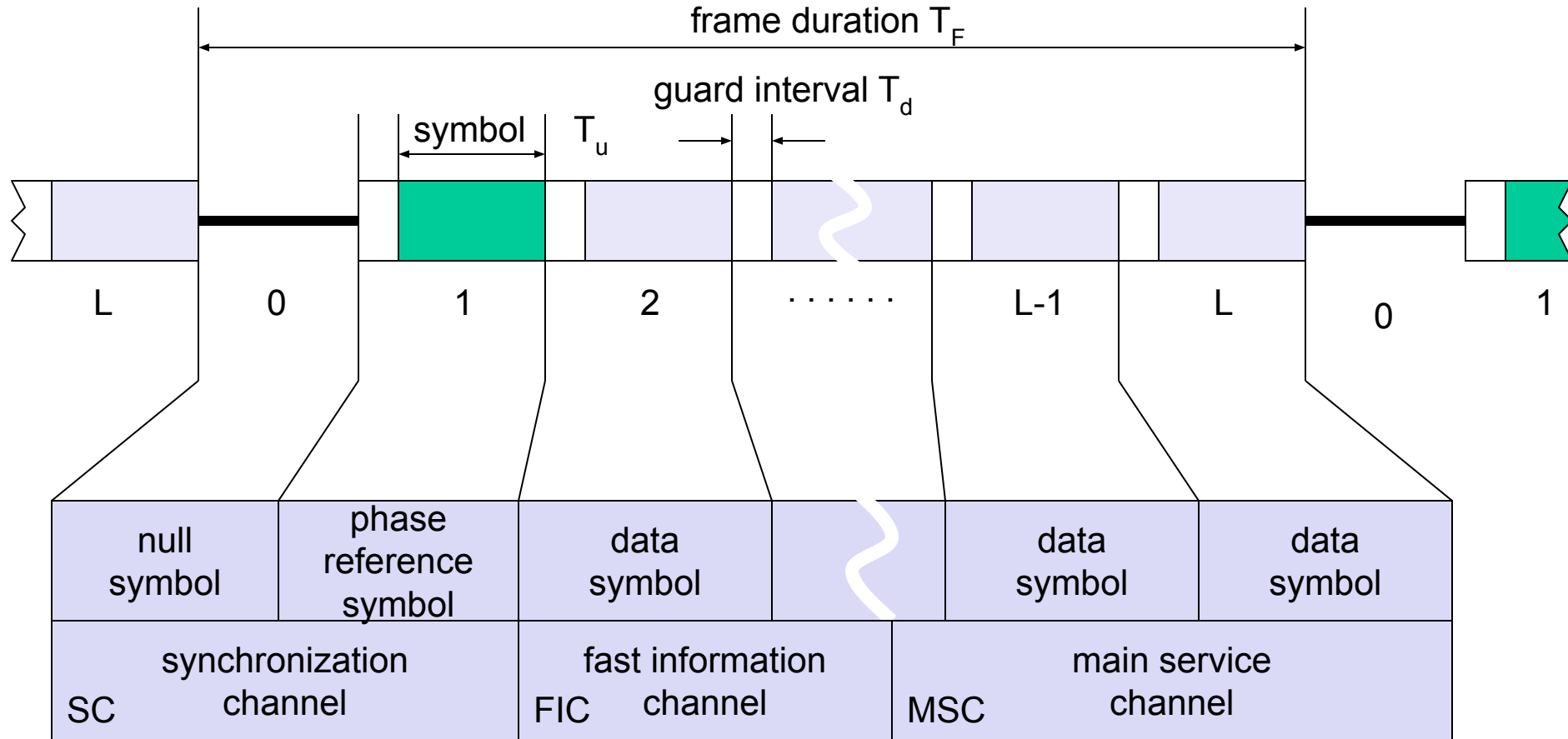
Packet mode

- ❑ transfer addressable packets



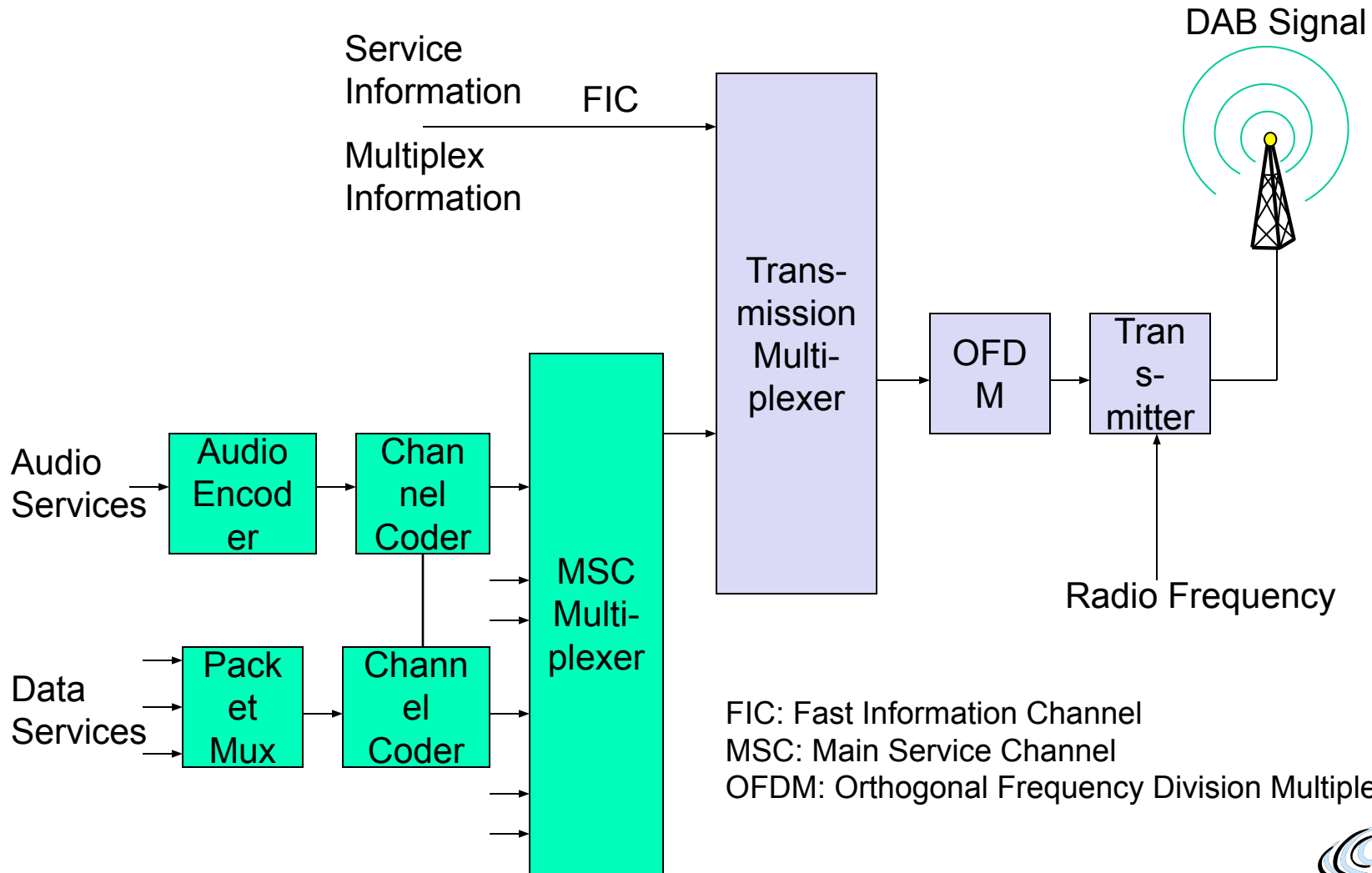


Transmission frame



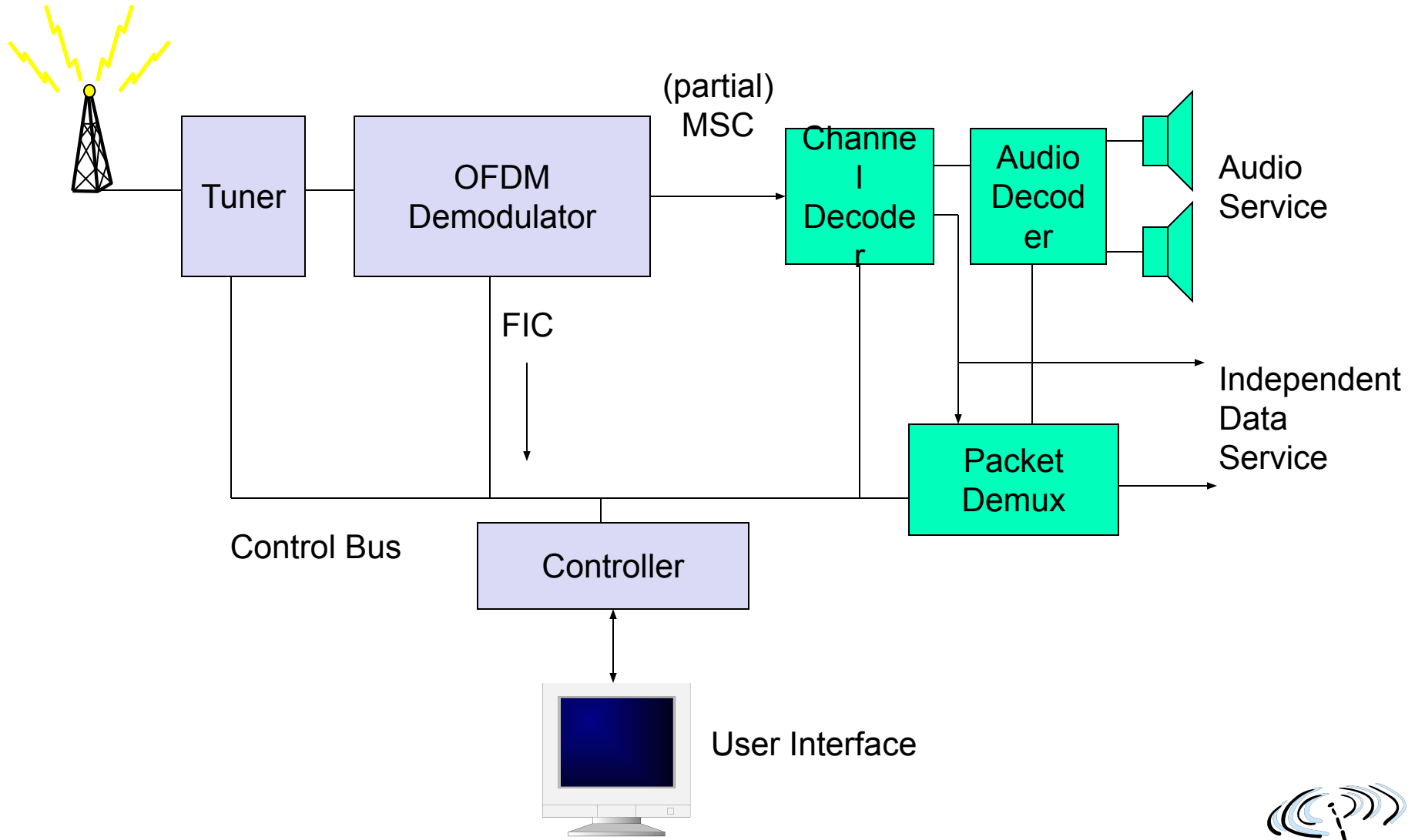


DAB sender





DAB receiver





Audio coding

- ❑ Goal
 - ❑ audio transmission almost with CD quality
 - ❑ robust against multipath propagation
 - ❑ minimal distortion of audio signals during signal fading
- ❑ Mechanisms
 - ❑ fully digital audio signals (PCM, 16 Bit, 48 kHz, stereo)
 - ❑ MPEG compression of audio signals, compression ratio 1:10
 - ❑ redundancy bits for error detection and correction
 - ❑ burst errors typical for radio transmissions, therefore signal interleaving - receivers can now correct single bit errors resulting from interference
 - ❑ low symbol-rate, many symbols
 - transmission of digital data using long symbol sequences, separated by guard spaces
 - delayed symbols, e.g., reflection, still remain within the guard space





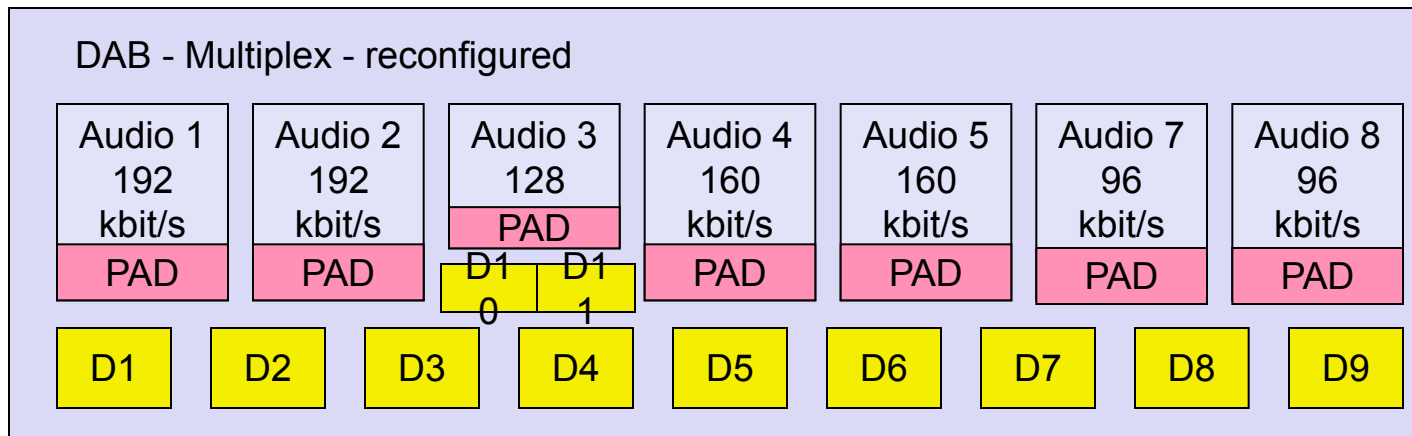
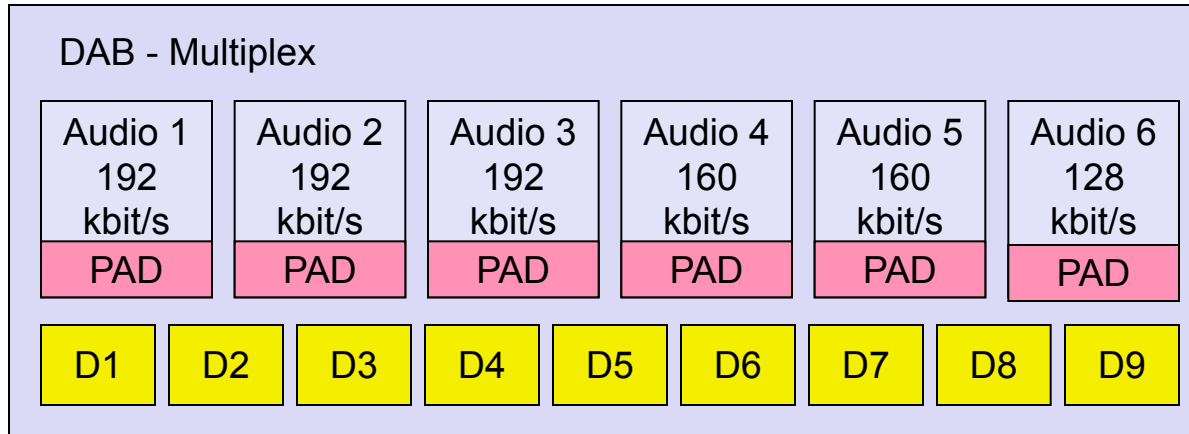
Bit rate management

- ❑ a DAB ensemble combines audio programs and data services with different requirements for transmission quality and bit rates
- ❑ the standard allows dynamic reconfiguration of the DAB multiplexing scheme (i.e., during transmission)
- ❑ data rates can be variable, DAB can use free capacities for other services
- ❑ the multiplexer performs this kind of bit rate management, therefore, additional services can come from different providers





Example of a reconfiguration





Multimedia Object Transfer Protocol (MOT)

Problem

- ❑ broad range of receiver capabilities
audio-only devices with single/multiple line text display, additional color graphic display, PC adapters etc.
- ❑ different types of receivers should at least be able to recognize all kinds of program associated and program independent data and process some of it

Solution

- ❑ common standard for data transmission: MOT
- ❑ important for MOT is the support of data formats used in other multimedia systems (e.g., online services, Internet, CD-ROM)
- ❑ DAB can therefore transmit HTML documents from the WWW with very little additional effort





MOT structure

MOT formats

- ❑ MHEG, Java, JPEG, ASCII, MPEG, HTML, HTTP, BMP, GIF, ...

Header core

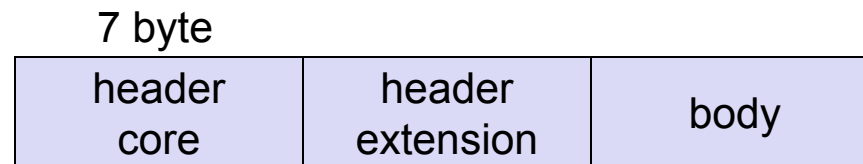
- ❑ size of header and body, content type

Header extension

- ❑ handling information, e.g., repetition distance, segmentation, priority
- ❑ information supports caching mechanisms

Body

- ❑ arbitrary data



DAB allows for many repetition schemes

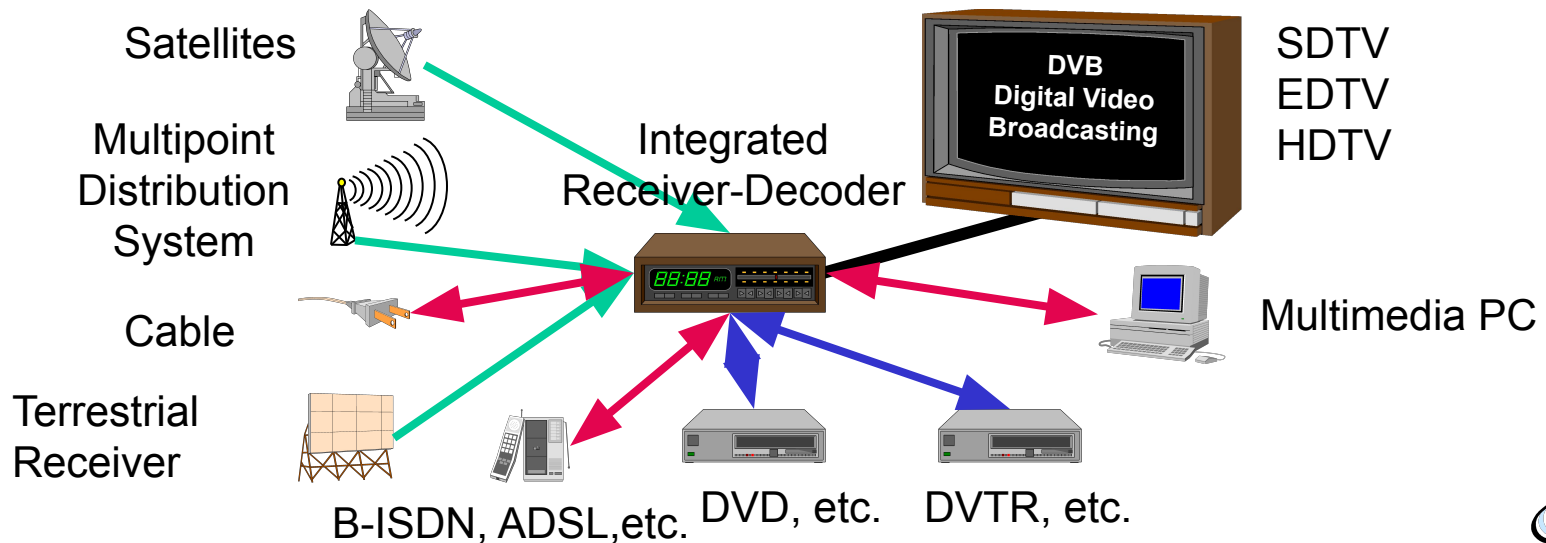
- ❑ objects, segments, headers





Digital Video Broadcasting

- ❑ 1991 foundation of the ELG (European Launching Group)
goal: development of digital television in Europe
- ❑ 1993 renaming into DVB (Digital Video Broadcasting)
goal: introduction of digital television based on
 - ❑ satellite transmission
 - ❑ cable network technology
 - ❑ later also terrestrial transmission

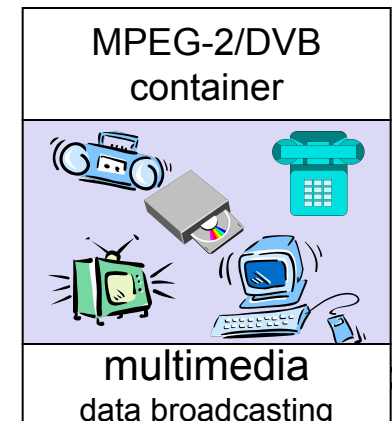
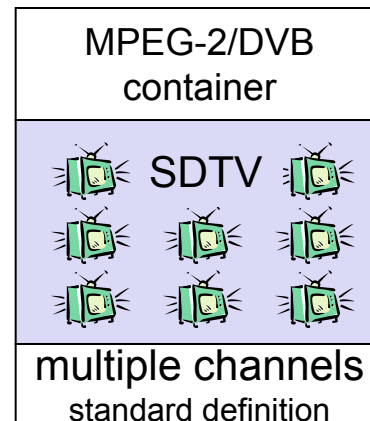
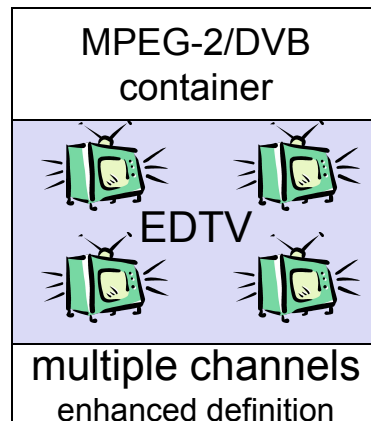
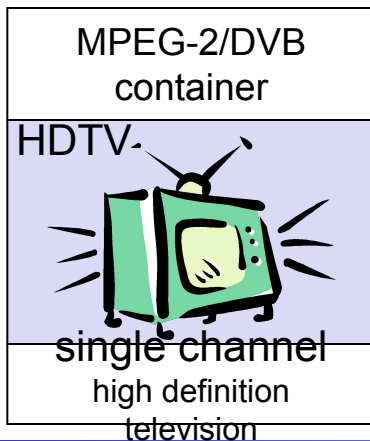




DVB Container

DVB transmits MPEG-2 container

- ❑ high flexibility for the transmission of digital data
- ❑ no restrictions regarding the type of information
- ❑ DVB Service Information specifies the content of a container
 - NIT (Network Information Table): lists the services of a provider, contains additional information for set-top boxes
 - SDT (Service Description Table): list of names and parameters for each service within a MPEG multiplex channel
 - EIT (Event Information Table): status information about the current transmission, additional information for set-top boxes
 - TDT (Time and Date Table): Update information for set-top boxes





Example: high-speed Internet

Asymmetric data exchange

- ❑ downlink: DVB receiver, data rate per user 6-38 Mbit/s
- ❑ return channel from user to service provider: e.g., modem with 33 kbit/s, ISDN with 64 kbit/s, ADSL with several 100 kbit/s etc.

